

Linear Algebra Done Right

Reading Notes

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1 Vector Spaces

1.1 $\mathbb{R}^n$ and $\mathbb{C}^n$

Before defining vector spaces, Axler establishes the fields we are working over. The symbol $\mathbb{F}$ will always denote either the real numbers $\mathbb{R}$ or the complex numbers $\mathbb{C}$.

Definition 1.1 (Fields). *A field is a set containing at least two distinct elements called 0 and 1, along with operations of addition and multiplication satisfying the properties of commutativity, associativity, identities, additive inverse, multiplicative inverse and distributivity.*

1.1.1 Complex Numbers, $\mathbb{C}$

Definition 1.2 (Complex Numbers).

- A complex number is an ordered pair (a, b) where $a, b \in \mathbb{R}$, but we will write this as $a + bi$.
- The set of all complex numbers is denoted by $\mathbb{C}$:

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}.$$

- Addition and multiplication on $\mathbb{C}$ are defined by

$$(a + bi) + (c + di) = (a + c) + (b + d)i,$$

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i;$$

here $a, b, c, d \in \mathbb{R}$

My thoughts

Just assume $i^2 = -1$ and everything works the same as in $\mathbb{R}$. Commutativity, associativity, identities, additive inverse, multiplicative inverse and distributivity are all properties of complex arithmetic.

1.1.2 Lists

Definition 1.3 (List, length).

- Suppose n is a nonnegative integer. A list of length n is an ordered collection of n elements (which might be numbers, other lists, or more abstract objects).
- Two lists are equal if and only if they have the same length and the same elements in the same order.

My thoughts

A list can have multiple same entries unlike sets. Fields and Vector Spaces (defined later) are sets, but a list is a tuple.

1.1.3 $\mathbb{F}^n$

Definition 1.4 ($\mathbb{F}^n$, coordinate). $\mathbb{F}^n$ is the set of all lists of length n of elements of $\mathbb{F}$:

$$\mathbb{F}^n = \{(x_1, \dots, x_n) : x_k \in \mathbb{F} \text{ for } k \in \{1, \dots, n\}\}$$

For $(x_1, \dots, x_n) \in \mathbb{F}^n$ and $k \in \{1, \dots, n\}$, we say that x_k is the k^{th} coordinate of $(x_1, \dots, x_n)$.

Definition 1.5 (addition in F^n). *Addition in F^n is defined by adding corresponding coordinates:*

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n)$$

Definition 1.6 (additive inverse in $\mathbb{F}^n$, $-x$). *For $x \in \mathbb{F}^n$, the additive inverse of x , denoted by $-x$, is the vector $-x \in \mathbb{F}^n$ such that*

$$x + (-x) = 0$$

Thus if $x = (x_1, \dots, x_n)$, then $-x = (-x_1, \dots, -x_n)$.

Definition 1.7 (scalar multiplication in $\mathbb{F}^n$). *The product of a number λ and a vector in $\mathbb{F}^n$ is computed by multiplying each coordinate of the vector by λ :*

$$\lambda(x_1, \dots, x_n) = (\lambda x_1, \dots, \lambda x_n);$$

here $\lambda \in \mathbb{F}$ and $(x_1, \dots, x_n) \in \mathbb{F}^n$

1.2 Definition of a Vector Space

Now we abstract the properties of $\mathbb{R}^n$ and $\mathbb{C}^n$ into a general structure.

Definition 1.8 (Vector Space). *A vector space is a set $\mathcal{V}$ along with an addition on $\mathcal{V}$ and a scalar multiplication on $\mathcal{V}$ such that commutativity, associativity, identities, additive inverse, and distributivity hold.*

My thoughts

Essentially, the difference between a field and a vector space is the fact that only scalar multiplication is defined for elements of a vector space. This also implies it is not necessary for a vector space to have multiplicative inverses. Also note, a field is a standalone set while a vector space relies on an underlying field to exist at all.

Definition 1.9 (Real Vector Space, Complex Vector Space).

- *A vector space over $\mathbb{R}$ is called a real vector space.*
- *A vector space over $\mathbb{C}$ is called a complex vector space.*

1.2.1 Some Property Proofs

Theorem 1.10 (Unique Additive Identity). *A vector space has a unique additive identity.*

Proof. Suppose 0 and $0'$ are both additive identities for V . Then

$$0' = 0' + 0 = 0 + 0' = 0,$$

where the first equality holds because 0 is an additive identity, the second equality comes from commutativity, and the third equality holds because $0'$ is an additive identity. Thus $0' = 0$, proving that $\mathcal{V}$ has only one additive identity. $\square$

Theorem 1.11. *a number times the vector 0 is 0*

Proof. For $v \in \mathcal{V}$, we have $0v = (0 + 0)v = 0v + 0v$. Adding the additive inverse of $0v$ to both sides of the equation above gives $0 = 0v$, as desired. $\square$

Theorem 1.12. *the number -1 times a vector gives the additive inverse of the vector*

Proof. For $v \in \mathcal{V}$, we have $v + (-1)v = 1v + (-1)v = (1 + (-1))v = 0v = 0$. This equation says that $(-1)v$, when added to v , gives 0 . Thus $(-1)v$ is the additive inverse of v , as desired $\square$

1.3 Subspaces

Definition 1.13 (Subspace). A subset U of $\mathcal{V}$ is called a subspace of $\mathcal{V}$ if U is also a vector space under the same addition and scalar multiplication.

1.3.1 Sums of Subspaces

Definition 1.14 (Sum of Subspaces). Suppose $\mathcal{V}_1, \dots, \mathcal{V}_m$ are subspaces of $\mathcal{V}$. The sum of $\mathcal{V}_1, \dots, \mathcal{V}_m$, denoted by $\mathcal{V}_1 + \dots + \mathcal{V}_m$, is the set of all possible sums of elements of $\mathcal{V}_1, \dots, \mathcal{V}_m$. More precisely,

$$\mathcal{V}_1 + \dots + \mathcal{V}_m = \{v_1 + \dots + v_m : v_1 \in \mathcal{V}_1, \dots, v_m \in \mathcal{V}_m\}$$

Theorem 1.15. Suppose $\mathcal{V}_1, \dots, \mathcal{V}_m$ are subspaces of $\mathcal{V}$. Then $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is the smallest subspace of $\mathcal{V}$ containing $\mathcal{V}_1, \dots, \mathcal{V}_m$.

My thoughts

I will add my own proof that sum of subspaces is a subspace (although Axler skips this) because I think it is necessary.

Proof. Let us first prove that sum of subspaces is a subspace

- $\mathcal{V}_1 + \dots + \mathcal{V}_m$ contains 0 (additive identity) since all the subspaces contain 0 and sum of subspaces contain every possible sum of their elements.
- Let $x, y \in (\mathcal{V}_1 + \dots + \mathcal{V}_m)$.
By definition we can express them as $x = v_1 + \dots + v_m$ and $y = u_1 + \dots + u_m$ such that $u_1, v_1 \in \mathcal{V}_1, \dots, u_m, v_m \in \mathcal{V}_m$. So we have

$$x + y = (v_1 + \dots + v_m) + (u_1 + \dots + u_m)$$

By commutativity and associativity of vector addition,

$$x + y = (v_1 + u_1) + \dots + (v_m + u_m)$$

Since $\mathcal{V}_1, \dots, \mathcal{V}_m$ are subspaces, $(v_1 + u_1) \in \mathcal{V}_1, \dots, (v_m + u_m) \in \mathcal{V}_m$

Hence we conclude $(x + y) \in \mathcal{V}_1 + \dots + \mathcal{V}_m$

Thus, it is closed under addition

- Let $x \in (\mathcal{V}_1 + \dots + \mathcal{V}_m)$. and $x = v_1 + \dots + v_m$ where $v_1 \in \mathcal{V}_1, \dots, v_m \in \mathcal{V}_m$. Let c be any scalar from the field $\mathbb{F}$.
Multiplying x with the scalar c yields:

$$cx = c(v_1 + \dots + v_m) = cv_1 + \dots + cv_m$$

Since $\mathcal{V}_1 + \dots + \mathcal{V}_m$ are subspaces $cv_1 \in \mathcal{V}_1, \dots, cv_m \in \mathcal{V}_m$.

We conclude $cx \in \mathcal{V}_1 + \dots + \mathcal{V}_m$

Thus, sum of subspaces is a subspace.

Now we can start with the proof for $\mathcal{V}_1, \dots, \mathcal{V}_m$ being the smallest subspace of $\mathcal{V}$ containing $\mathcal{V}_1, \dots, \mathcal{V}_m$.

The subspaces $\mathcal{V}_1, \dots, \mathcal{V}_m$ are all contained in $\mathcal{V}_1 + \dots + \mathcal{V}_m$, considering sums $v_1 + \dots + v_m$ where all except one of the v_k 's are 0.

Conversely, every subspace of $\mathcal{V}$ containing $\mathcal{V}_1, \dots, \mathcal{V}_m$ contains $\mathcal{V}_1 + \dots + \mathcal{V}_m$ (because subspaces must contain all finite sums of their elements). Thus $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is the smallest subspace of $\mathcal{V}$ containing $\mathcal{V}_1, \dots, \mathcal{V}_m$. $\square$

1.3.2 Direct Sums

Definition 1.16 (Direct Sum). Suppose $\mathcal{V}_1, \dots, \mathcal{V}_m$ are subspaces of $\mathcal{V}$.

- The sum $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is called a *direct sum* if each element of $\mathcal{V}_1 + \dots + \mathcal{V}_m$ can be written in only one way as a sum $v_1 + \dots + v_m$, where each $v_k \in \mathcal{V}_k$.
- If $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is a direct sum, then $\mathcal{V}_1 \oplus \dots \oplus \mathcal{V}_m$ denotes $\mathcal{V}_1 + \dots + \mathcal{V}_m$, with the $\oplus$ notation serving as an indication that this is a direct sum.

Theorem 1.17. Suppose $\mathcal{V}_1 + \dots + \mathcal{V}_m$ are subspaces of $\mathcal{V}$. Then $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is a direct sum if and only if the only way to write 0 as a sum $v_1 + \dots + v_m$, where each $v_k \in \mathcal{V}_k$, is by taking each v_k equal to 0.

Proof. First suppose $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is a direct sum. Then the definition of direct sum implies that the only way to write 0 as a sum $v_1 + \dots + v_m$, where each $v_k \in \mathcal{V}_k$, is by taking each v_k equal to 0.

My thoughts

This is because taking every $v_k = 0$ is a trivial solution, but since we know it is a direct sum, the only solution would be taking every $v_k = 0$.

Now suppose that the only way to write 0 as a sum $v_1 + \dots + v_m$, where each $v_k \in \mathcal{V}_k$, is by taking each v_k equal to 0. To show that $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is a direct sum, let $v \in \mathcal{V}_1 + \dots + \mathcal{V}_m$. We can write

$$v = v_1 + \dots + v_m$$

for some $v_1 \in \mathcal{V}_1, \dots, v_m \in \mathcal{V}_m$. To show that this representation is unique, suppose we also have

$$v = u_1 + \dots + u_m$$

where $u_1 \in \mathcal{V}_1, \dots, u_m \in \mathcal{V}_m$. Subtracting these two equations, we have

$$0 = (v_1 - u_1) + \dots + (v_m - u_m)$$

Because $v_1 - u_1 \in \mathcal{V}_1, \dots, v_m - u_m \in \mathcal{V}_m$, the equation above implies that each $v_k - u_k$ equals 0. Thus $v_1 = u_1, \dots, v_m = u_m$, as desired. $\square$

Theorem 1.18. Suppose U and W are subspaces of V . Then

$$U + W \text{ is a direct sum} \iff U \cap W = \{0\}.$$

Proof. First suppose that $U + W$ is a direct sum. If $v \in U \cap W$, then $0 = v + (-v)$, where $v \in U$ and $-v \in W$. By the unique representation of 0 as the sum of a vector in U and a vector in W , we have $v = 0$. Thus $U \cap W = \{0\}$, completing the proof in one direction. To prove the other direction, now suppose $U \cap W = \{0\}$. To prove that $U + W$ is a direct sum, suppose $u \in U, w \in W$, and $0 = u + w$. To complete the proof, we only need to show that $u = w = 0$. The equation above implies that $u = -w \in W$. Thus $u \in U \cap W$. Hence $u = 0$, which by the equation above implies that $w = 0$, completing the proof. $\square$

1.4 Exercise 1A

My thoughts

I can't find a way to prove these identities by themselves, maybe I need to learn analysis for that. I will just be assuming $\mathbb{R}$ is already known to be a field, and use those properties to prove these.

Question 1. Show that $\alpha + \beta = \beta + \alpha$ for all $\alpha, \beta \in \mathbb{C}$.

Proof. Letting $\alpha = a + bi$, $\beta = c + di$, $\alpha + \beta = a + ib + c + id$

$$\begin{aligned} &= (a + c) + (b + d)i \\ &= (c + a) + (d + b)i \\ &= (c + id) + (a + ib) \\ &= \beta + \alpha \end{aligned}$$

□

Question 2. Show that $(\alpha + \beta) + \lambda = \alpha + (\beta + \lambda)$ for all $\alpha, \beta, \lambda \in \mathbb{C}$.

Proof. Letting $\alpha = a + bi$, $\beta = c + di$, and $\lambda = e + fi$ (where $a, b, c, d, e, f \in \mathbb{R}$), the left side of the equation expands as:

$$\begin{aligned} &(\alpha + \beta) + \lambda \\ &= ((a + c) + (b + d)i) + (e + fi) \\ &= ((a + c) + e) + ((b + d) + f)i \end{aligned}$$

Because addition is associative in the real numbers, we can rewrite the real and imaginary components as $(a + (c + e))$ and $(b + (d + f))$.

Expanding the right side gives:

$$\begin{aligned} &\alpha + (\beta + \lambda) \\ &= (a + bi) + ((c + e) + (d + f)i) \\ &= (a + (c + e)) + (b + (d + f))i \end{aligned}$$

□

Question 3. Show that $(\alpha\beta)\lambda = \alpha(\beta\lambda)$ for all $\alpha, \beta, \lambda \in \mathbb{C}$.

Proof. Letting $\alpha = a + bi$, $\beta = c + di$, and $\lambda = e + fi$ (where $a, b, c, d, e, f \in \mathbb{R}$), the left side of the equation expands as:

$$((ac - bd) + (ad + bc)i)(e + fi)$$

Multiplying this out yields a real part of $ace - bde - adf - bcf$ and an imaginary part of $acf - bdf + ade + bce$.

Expanding the right side $\alpha(\beta\lambda)$ gives

$$(a + bi)((ce - df) + (cf + de)i)$$

Multiplying this out yields a real part of $ace - adf - bcf - bde$ and an imaginary part of $acf + ade + bce - bdf$. □

Question 4. Show that $\lambda(\alpha + \beta) = \lambda\alpha + \lambda\beta$ for all $\lambda, \alpha, \beta \in \mathbb{C}$

Proof. Expanding $\lambda(\alpha + \beta)$ with $\lambda = e + fi$, $\alpha = a + bi$, and $\beta = c + di$ gives a real part of $ea + ec - fb - fd$ and an imaginary part of $eb + ed + fa + fc$. Multiplying out $\lambda\alpha + \lambda\beta$ yields the exact same terms, relying on the distributive property of real numbers to match them up. □

Question 5. Show that for every $\alpha \in \mathbb{C}$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha + \beta = 0$.

Proof. Setting $\alpha + \beta = 0$ with $\alpha = a + bi$ and $\beta = c + di$ results in $(a + c) + (b + d)i = 0$. Equating the real and imaginary components to zero means $a + c = 0$ and $b + d = 0$, yielding exactly $c = -a$ and $d = -b$. Because every real number has a uniquely defined additive inverse in $\mathbb{R}$, the components c and d exist and are unique, making β unique. □

Question 6. Show that for every $\alpha \in \mathbb{C}$ with $\alpha \neq 0$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha\beta = 1$.

Proof. $\alpha = a + bi$, $\beta = c + di$

Setting $(a + bi)(c + di) = 1$ results in $(ac - bd) + (ad + bc)i = 1 + 0i$. Equating the real and imaginary parts gives a system of two linear equations: $ac - bd = 1$ and $ad + bc = 0$. Since $\alpha \neq 0$, we know that a and b are not both zero, meaning $a^2 + b^2 \neq 0$. Solving this system for c and d yields exactly $c = \frac{a}{a^2+b^2}$ and $d = \frac{-b}{a^2+b^2}$. This produces a single, well-defined real value for both c and d . $\square$

Question 7. Show that $\frac{-1+\sqrt{3}i}{2}$ is a cube root of 1 (meaning that its cube equals 1).

Proof. Squaring $\frac{-1+\sqrt{3}i}{2}$ yields $\frac{-1-\sqrt{3}i}{2}$. Multiplying that squared result by the original term $\frac{-1+\sqrt{3}i}{2}$ gives $\frac{(-1)^2 - (\sqrt{3}i)^2}{4} = \frac{1 - (-3)}{4}$, which reduces perfectly to 1. $\square$

Question 8. Find two distinct square roots of i .

Proof. $\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$ and $-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$ $\square$

Question 9. Find $x \in \mathbb{R}^4$ such that $(4, -3, 1, 7) + 2x = (5, 9, -6, 8)$.

Proof. $(1/2, 6, -7/2, 1/2)$ $\square$

Question 10. Explain why there does not exist $\lambda \in \mathbb{C}$ such that $\lambda(2 - 3i, 5 + 4i, -6 + 7i) = (12 - 5i, 7 + 22i, -32 - 9i)$.

Proof. Setting $\lambda(2 - 3i) = 12 - 5i$ for the first component gives $\lambda = \frac{12-5i}{2-3i}$.

Multiplying the numerator and denominator by the conjugate $(2 + 3i)$ yields $\frac{24+36i-10i+15}{13}$, which simplifies to $\lambda = 3 + 2i$.

Testing this λ on the second component yields $(3 + 2i)(5 + 4i) = 15 + 12i + 10i - 8 = 7 + 22i$, which matches the required vector perfectly.

Now, applying $\lambda = 3 + 2i$ to the third component gives $(3 + 2i)(-6 + 7i) = -18 + 21i - 12i - 14 = -32 + 9i$. But the required third component in the target vector is $-32 - 9i$. $\square$

Question 11. Show that $(x + y) + z = x + (y + z)$ for all $x, y, z \in \mathbb{F}^n$.

Proof. Let $x = (x_1, \dots, x_n)$, $y = (y_1, \dots, y_n)$, and $z = (z_1, \dots, z_n)$. Adding component-wise, the left side gives $((x_1 + y_1) + z_1, \dots, (x_n + y_n) + z_n)$. Because addition is associative in the underlying field $\mathbb{F}$, each component $(x_i + y_i) + z_i$ is equal to $x_i + (y_i + z_i)$. This perfectly constructs the right side $x + (y + z)$. $\square$

Question 12. Show that $(ab)x = a(bx)$ for all $x \in \mathbb{F}^n$ and all $a, b \in \mathbb{F}$.

Proof. Expanding x as $(x_1, \dots, x_n)$ means the left side $(ab)x$ becomes $((ab)x_1, \dots, (ab)x_n)$. The right side $a(bx)$ evaluates to $a(bx_1, \dots, bx_n)$, which then scales to $(a(bx_1), \dots, a(bx_n))$. Because multiplication within the underlying field $\mathbb{F}$ is associative, $(ab)x_i = a(bx_i)$ for every component $\square$

Question 13. Show that $1x = x$ for all $x \in \mathbb{F}^n$.

Proof. Expanding x into its components $(x_1, \dots, x_n)$ gives $1x = (1x_1, \dots, 1x_n)$. Because 1 is the multiplicative identity in the field $\mathbb{F}$, $1x_i = x_i$ for every individual component. This gives us the original vector x . $\square$

Question 14. Show that $\lambda(x + y) = \lambda x + \lambda y$ for all $\lambda \in \mathbb{F}$ and all $x, y \in \mathbb{F}^n$.

Proof. Expanding $\lambda(x + y)$ component-wise gives $(\lambda(x_1 + y_1), \dots, \lambda(x_n + y_n))$. Because the underlying field $\mathbb{F}$ has the distributive property, each component expands to $\lambda x_i + \lambda y_i$. This perfectly matches the component-wise expansion of the right side, $\lambda x + \lambda y = (\lambda x_1 + \lambda y_1, \dots, \lambda x_n + \lambda y_n)$. $\square$

Question 15. Show that $(a + b)x = ax + bx$ for all $a, b \in \mathbb{F}$ and all $x \in \mathbb{F}^n$.

Proof. Expanding $(a + b)x$ component-wise yields $((a + b)x_1, \dots, (a + b)x_n)$. Applying the distributive property of the underlying field $\mathbb{F}$ to each component gives $(ax_1 + bx_1, \dots, ax_n + bx_n)$. This perfectly matches the right side $ax + bx$ when you add the scaled vectors $(ax_1, \dots, ax_n)$ and $(bx_1, \dots, bx_n)$ together. $\square$

1.5 Exercise 1B

Question 1. Prove that $-(-v) = v$ for every $v \in \mathcal{V}$.

Proof.

$$\begin{aligned} -(-v) &= -(-v) + 0 \\ -(-v) &= -(-v) + (-v + v) \\ -(-v) &= (-(-v) + -v) + v \\ -(-v) &= 0 + v \\ -(-v) &= v \end{aligned}$$

$\square$

Question 2. Suppose $a \in \mathbb{F}, v \in \mathcal{V}$, and $av = 0$. Prove that $a = 0$ or $v = 0$.

Proof. Given equation:

$$av = 0$$

Since $a \in \mathbb{F}$ (a field) and $a \neq 0$, it has a multiplicative inverse $a^{-1} \in \mathbb{F}$.

$$\begin{aligned} a^{-1}(av) &= a^{-1}(0) \\ a^{-1}(av) &= 0 \end{aligned}$$

By associativity of scalar multiplication:

$$\begin{aligned} (a^{-1}a)v &= 0 \\ 1v &= 0 \\ v &= 0 \end{aligned}$$

If $a \neq 0$, then v must be 0. Therefore, if $av = 0$, it must be true that either $a = 0$ or $v = 0$. $\square$

Question 3. Suppose $v, w \in \mathcal{V}$. Explain why there exists a unique $x \in \mathcal{V}$ such that $v + 3x = w$.

Proof. First we prove existence:

$$x = \frac{1}{3}(w - v)$$

Since a vector space is closed under addition and scalar multiplication $x \in \mathcal{V}$

Now for uniqueness:

Suppose there are two vectors, x_1 and x_2 in V , that both satisfy the equation.

$$\begin{aligned} v + 3x_1 &= v + 3x_2 \\ -v + (v + 3x_1) &= -v + (v + 3x_2) \end{aligned}$$

Using associativity and additive inverse properties,

$$\begin{aligned} 3x_1 &= 3x_2 \\ x_1 &= x_2 \end{aligned}$$

Thus x is unique. $\square$

Question 4. The empty set is not a vector space. The empty set fails to satisfy only one of the requirements listed in the definition of a vector space. Which one?

Proof. It doesn't have the zero element. $\square$

Question 5. Show that in the definition of a vector space, the additive inverse condition can be replaced with the condition that $0v = 0$ for all $v \in \mathcal{V}$.

Proof. Let's assume the additive inverse condition does not exist. Instead we have $0v = 0$ for all $v \in \mathcal{V}$. We can arrive at the additive inverse criterion from here.

$$0v = 0$$

$$(1 + (-1))v = 0$$

Using distributivity,

$$1v + (-1)v = 0$$

Using multiplicative identity,

$$v + (-1)v = 0$$

The element $(-1)v$ satisfies the exact requirements of an additive inverse. $\square$

Question 6. Let ∞ and $-\infty$ denote two distinct objects, neither of which is in $\mathbb{R}$. Define an addition and scalar multiplication on $\mathbb{R} \cup \{\infty, -\infty\}$ as you could guess from the notation. Specifically, the sum and product of two real numbers is as usual, and for $t \in \mathbb{R}$ define

$$t\infty = \begin{cases} -\infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ \infty & \text{if } t > 0, \end{cases} \quad t(-\infty) = \begin{cases} \infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ -\infty & \text{if } t > 0, \end{cases}$$

and

$$\begin{aligned} t + \infty &= \infty + t = \infty + \infty = \infty, \\ t + (-\infty) &= (-\infty) + t = (-\infty) + (-\infty) = -\infty, \\ \infty + (-\infty) &= (-\infty) + \infty = 0. \end{aligned}$$

With these operations of addition and scalar multiplication, is $\mathbb{R} \cup \{\infty, -\infty\}$ a vector space over $\mathbb{R}$? Explain.

Proof. It is not a vector space.

If $t + \infty = \infty$, we have:

$$\begin{aligned} 1 + \infty &= 2 + \infty \\ 1 &= 2 \end{aligned}$$

We can cancel out the ∞ on both sides due to existence of additive inverse. $\square$

Question 7. Suppose S is a nonempty set. Let $\mathcal{V}^S$ denote the set of functions from S to $\mathcal{V}$. Define a natural addition and scalar multiplication on $\mathcal{V}^S$, and show that $\mathcal{V}^S$ is a vector space with these definitions.

Proof. Because $\mathcal{V}$ is a vector space, $\mathcal{V}^S$ will obey all vector space axioms through point-wise evaluations for every $x \in S$.

- Commutativity & Associativity: $(f + g)(x) = f(x) + g(x) = g(x) + f(x) = (g + f)(x)$. Associativity follows the exact same logic.

- Additive Identity: The zero function, defined by $0(x) = 0$ for all $x \in S$.
- Additive Inverse: The function defined point-wise by $(-f)(x) = -(f(x))$.
- Distributive Properties: $a(f + g)(x) = a(f(x) + g(x)) = af(x) + ag(x) = (af + ag)(x)$.

□

Question 8. Suppose $\mathcal{V}$ is a real vector space.

- The *complexification* of $\mathcal{V}$, denoted by $\mathcal{V}_{\mathbb{C}}$, equals $\mathcal{V} \times \mathcal{V}$. An element of $\mathcal{V}_{\mathbb{C}}$ is an ordered pair (u, v) , where $u, v \in \mathcal{V}$, but we write this as $u + iv$.
- Addition on $\mathcal{V}_{\mathbb{C}}$ is defined by

$$(u_1 + iv_1) + (u_2 + iv_2) = (u_1 + u_2) + i(v_1 + v_2)$$

for all $u_1, v_1, u_2, v_2 \in \mathcal{V}$.

- Complex scalar multiplication on $\mathcal{V}_{\mathbb{C}}$ is defined by

$$(a + bi)(u + iv) = (au - bv) + i(av + bu)$$

for all $a, b \in \mathbb{R}$ and all $u, v \in \mathcal{V}$.

Prove that with the definitions of addition and scalar multiplication as above, $\mathcal{V}_{\mathbb{C}}$ is a complex vector space.

Proof. Because $\mathcal{V}$ is a real vector space, it is inherently closed under its own addition and scalar multiplication. Therefore, $(u_1 + u_2) \in \mathcal{V}$, $(v_1 + v_2) \in \mathcal{V}$, $(au - bv) \in \mathcal{V}$, and $(av + bu) \in \mathcal{V}$, guaranteeing the operations always produce valid elements of $\mathcal{V}_{\mathbb{C}}$.

The following properties are also satisfied:

- Additive Identity: $0 + i0$ (since $0 \in \mathcal{V}$).
- Additive Inverse: $-(u) + i(-v)$ (since $\mathcal{V}$ contains inverses for u and v).
- Multiplicative Identity: $1 + i0$ acts as the scalar 1, yielding $1u - 0v + i(1v + 0u) = u + iv$.
- Associativity, Commutativity, and Distributivity: These expand algebraically and hold true because the vectors in $\mathcal{V}$ and scalars in $\mathbb{R}$ already obey these rules.

□

1.6 Exercise 1C

Question 1. For each of the following subsets of $\mathbf{F}^3$, determine whether it is a subspace of $\mathbf{F}^3$.

- $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$
- $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$
- $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1x_2x_3 = 0\}$
- $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$

Proof. We only need to check for closure under addition and scalar multiplication and existence of a zero element. By this logic, (a),(d) are subspaces while (b),(c) are not. □

Question 2. Verify all assertions about subspaces below

- (a) If $b \in \mathbb{F}$, then

$$\{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 : x_3 = 5x_4 + b\}$$

is a subspace of $\mathbb{F}^4$ if and only if $b = 0$.

- (b) The set of continuous real-valued functions on the interval $[0, 1]$ is a subspace of $\mathbb{R}^{[0,1]}$.
 (c) The set of differentiable real-valued functions on $\mathbb{R}$ is a subspace of $\mathbb{R}^{\mathbb{R}}$.
 (d) The set of differentiable real-valued functions f on the interval $(0, 3)$ such that $f'(2) = b$ is a subspace of $\mathbb{R}^{(0,3)}$ if and only if $b = 0$.
 (e) The set of all sequences of complex numbers with limit 0 is a subspace of $\mathbb{C}^{\infty}$.

Proof. (a) Forward direction (If it is a subspace, then $b = 0$): Every subspace must contain the additive identity, which in $\mathbb{F}^4$ is the zero vector $(0, 0, 0, 0)$. Substituting these coordinates into the defining equation yields $0 = 5(0) + b$, which simplifies strictly to $b = 0$.

Reverse direction (If $b = 0$, then it is a subspace): Setting $b = 0$ changes the subset condition to $x_3 = 5x_4$. Clearly, this is closed under addition, scalar multiplication and the zero vector $(0, 0, 0, 0)$ exists.

- (b) To prove that W is a subspace of $\mathbb{R}^{[0,1]}$, we must verify the three conditions of the Subspace Criterion:

- (a) Presence of the zero vector: The zero vector in $\mathbb{R}^{[0,1]}$ is the zero function, denoted by $\mathbf{0}$, where $\mathbf{0}(x) = 0$ for all $x \in [0, 1]$. Because constant functions are continuous on $\mathbb{R}$, the zero function is continuous on $[0, 1]$. Therefore, $\mathbf{0} \in W$, proving that W is non-empty.
 (b) Closure under vector addition: Let $f, g \in W$. By definition, f and g are both continuous functions on $[0, 1]$. From the properties of continuous functions in real analysis, the sum of two continuous functions is continuous. Therefore, the function $(f + g)(x) = f(x) + g(x)$ is continuous on $[0, 1]$, which implies $f + g \in W$.
 (c) Closure under scalar multiplication: Let $f \in W$ and $c \in \mathbb{R}$. Since f is a continuous function on $[0, 1]$, its multiplication by a scalar constant c results in a continuous function. Thus, the function $(cf)(x) = cf(x)$ is continuous on $[0, 1]$, meaning $cf \in W$.

Since W contains the zero vector and is closed under both addition and scalar multiplication, W is a subspace of $\mathbb{R}^{[0,1]}$.

- (c) To prove that W is a subspace of $\mathbb{R}^{\mathbb{R}}$, we verify the three conditions of the Subspace Criterion:

- (a) Presence of the zero vector: The zero vector in $\mathbb{R}^{\mathbb{R}}$ is the zero function, denoted by $\mathbf{0}$, where $\mathbf{0}(x) = 0$ for all $x \in \mathbb{R}$. Because constant functions are differentiable everywhere on $\mathbb{R}$ (with a derivative of 0), the zero function is differentiable. Therefore, $\mathbf{0} \in W$, proving that W is non-empty.
 (b) Closure under vector addition: Let $f, g \in W$. By definition, f and g are both differentiable functions on $\mathbb{R}$. From standard calculus theorems, the sum of two differentiable functions is differentiable, and its derivative is given by $(f + g)'(x) = f'(x) + g'(x)$. Therefore, the function $(f + g)$ is differentiable on $\mathbb{R}$, which implies $f + g \in W$.
 (c) Closure under scalar multiplication: Let $f \in W$ and $c \in \mathbb{R}$. Since f is a differentiable function on $\mathbb{R}$, its multiplication by a scalar constant c results in a differentiable function, with the derivative $(cf)'(x) = cf'(x)$. Thus, the function cf is differentiable on $\mathbb{R}$, meaning $cf \in W$.

Since W contains the zero vector and is closed under both vector addition and scalar multiplication, W is a subspace of $\mathbb{R}^{\mathbb{R}}$.

(d) Let $\mathcal{V} = \mathbb{R}^{(0,3)}$ and let $W_b = \{f \in \mathcal{V} \mid f \text{ is differentiable on } (0,3) \text{ and } f'(2) = b\}$. We must prove that W_b is a subspace of $\mathcal{V}$ if and only if $b = 0$.

($\Rightarrow$) Assume W_b is a subspace of $\mathcal{V}$. Then it must contain the zero vector of $\mathcal{V}$, which is the zero function $\mathbf{0}$ defined by $\mathbf{0}(x) = 0$ for all $x \in (0,3)$. The derivative of the zero function is $\mathbf{0}'(x) = 0$ for all $x \in (0,3)$. Since $\mathbf{0} \in W_b$, it must satisfy the condition defining the set W_b , which means $\mathbf{0}'(2) = b$. Thus, $0 = b$.

($\Leftarrow$) Assume $b = 0$. We must show that $W_0 = \{f \in \mathcal{V} \mid f \text{ is differentiable on } (0,3) \text{ and } f'(2) = 0\}$ is a subspace. We verify the three conditions of the subspace criterion:

1. Presence of the zero vector: The zero function $\mathbf{0}$ is differentiable, and as shown above, $\mathbf{0}'(2) = 0$. Thus, $\mathbf{0} \in W_0$.
2. Closure under vector addition: Let $f, g \in W_0$. Both are differentiable on $(0,3)$, so their sum $f + g$ is also differentiable. By the linearity of the derivative, $(f + g)'(2) = f'(2) + g'(2) = 0 + 0 = 0$. Therefore, $f + g \in W_0$.
3. Closure under scalar multiplication: Let $f \in W_0$ and $c \in \mathbb{R}$. The function cf is differentiable, and its derivative at $x = 2$ is $(cf)'(2) = cf'(2) = c(0) = 0$. Therefore, $cf \in W_0$.

Since W_0 contains the zero vector and is closed under both addition and scalar multiplication, it is a subspace of $\mathcal{V}$.

Therefore, W_b is a subspace of $\mathbb{R}^{(0,3)}$ if and only if $b = 0$.

My thoughts

I don't know anything about complex sequences, but it is easy enough to wager a guess after a google search.

(e)

Let $\mathbb{C}^\infty$ be the vector space of all sequences of complex numbers, and let W be the set of all sequences of complex numbers with limit 0 [cite: 3]. We will prove that W is a subspace of $\mathbb{C}^\infty$.

We verify the three conditions of the subspace criterion:

1. Presence of the zero vector: The zero vector in $\mathbb{C}^\infty$ is the zero sequence, denoted by $\mathbf{0} = (0, 0, 0, \dots)$, where $z_n = 0$ for all $n \geq 1$. The limit of this constant sequence is $\lim_{n \rightarrow \infty} 0 = 0$. Therefore, $\mathbf{0} \in W$, meaning the subset is non-empty.
2. Closure under vector addition: Let $x = (x_n)$ and $y = (y_n)$ be sequences in W . By definition, $\lim_{n \rightarrow \infty} x_n = 0$ and $\lim_{n \rightarrow \infty} y_n = 0$. According to the algebraic limit theorems for sequences, the limit of their sum is given by $\lim_{n \rightarrow \infty} (x_n + y_n) = \lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n = 0 + 0 = 0$. Thus, the sequence $x + y \in W$.
3. Closure under scalar multiplication: Let $x = (x_n) \in W$ and let $\lambda \in \mathbb{C}$. By definition, $\lim_{n \rightarrow \infty} x_n = 0$. Using the limit laws for scalar multiples, $\lim_{n \rightarrow \infty} (\lambda x_n) = \lambda \lim_{n \rightarrow \infty} x_n = \lambda \cdot 0 = 0$. Therefore, the sequence $\lambda x \in W$.

So W is a subspace of $\mathbb{C}^\infty$. □

Question 3. Show that the set of differentiable real-valued functions f on the interval $(-4, 4)$ such that $f'(-1) = 3f(2)$ is a subspace of $\mathbf{R}^{(-4,4)}$.

Proof. Let $W = \{f \in V \mid f \text{ is differentiable on } (-4, 4) \text{ and } f'(-1) = 3f(2)\}$. We verify that W satisfies the three conditions of a subspace:

1. **Additive identity:** The zero function, denoted by $\mathbf{0}$, where $\mathbf{0}(x) = 0$ for all $x \in (-4, 4)$, is differentiable. Its derivative is everywhere zero, so $\mathbf{0}'(-1) = 0$. Evaluating the other side gives $3\mathbf{0}(2) = 3(0) = 0$. Thus, $\mathbf{0}'(-1) = 3\mathbf{0}(2)$, meaning $\mathbf{0} \in W$.
2. **Closed under addition:** Let $f, g \in W$. Since f and g are differentiable, their sum $f + g$ is also differentiable. By the linearity of the derivative, we have:

$$(f + g)'(-1) = f'(-1) + g'(-1)$$

Because $f, g \in W$, we can substitute $f'(-1) = 3f(2)$ and $g'(-1) = 3g(2)$:

$$(f + g)'(-1) = 3f(2) + 3g(2) = 3(f(2) + g(2)) = 3(f + g)(2)$$

Therefore, $f + g \in W$.

3. **Closed under scalar multiplication:** Let $f \in W$ and $\lambda \in \mathbb{R}$. The scalar multiple λf is differentiable, and by the linearity of the derivative:

$$(\lambda f)'(-1) = \lambda f'(-1)$$

Since $f \in W$, we know $f'(-1) = 3f(2)$, yielding:

$$(\lambda f)'(-1) = \lambda(3f(2)) = 3(\lambda f(2)) = 3(\lambda f)(2)$$

Therefore, $\lambda f \in W$.

Since W contains the zero vector and is closed under both addition and scalar multiplication, W is a subspace of $\mathbb{R}^{(-4,4)}$. $\square$

Question 4. Suppose $b \in \mathbb{R}$. Show that the set of continuous real-valued functions f on the interval $[0, 1]$ such that $\int_0^1 f = b$ is a subspace of $\mathbb{R}^{[0,1]}$ if and only if $b = 0$.

Proof. Let $W_b = \left\{ f \in V \mid f \text{ is continuous on } [0, 1] \text{ and } \int_0^1 f = b \right\}$. We want to show that W_b is a subspace of V if and only if $b = 0$.

($\Rightarrow$) Assume W_b is a subspace of $\mathbb{R}^{[0,1]}$. Then W_b must contain the additive identity of $\mathbb{R}^{[0,1]}$, which is the zero function $\mathbf{0}$ defined by $\mathbf{0}(x) = 0$ for all $x \in [0, 1]$. The zero function is continuous, and its integral is:

$$\int_0^1 \mathbf{0} = \int_0^1 0 \, dx = 0$$

Since $\mathbf{0} \in W_b$, it must satisfy the condition defining the set, meaning $\int_0^1 \mathbf{0} = b$. Therefore, $b = 0$.

($\Leftarrow$) Assume $b = 0$. We must show that $W_0 = \left\{ f \in \mathbb{R}^{[0,1]} \mid f \text{ is continuous on } [0, 1] \text{ and } \int_0^1 f = 0 \right\}$ is a subspace. We verify the three conditions:

1. **Additive identity:** As shown above, the zero function $\mathbf{0}$ is continuous and $\int_0^1 \mathbf{0} = 0$. Thus, $\mathbf{0} \in W_0$.
2. **Closed under addition:** Let $f, g \in W_0$. Since both are continuous, their sum $f + g$ is also continuous. By the linearity of the definite integral:

$$\int_0^1 (f + g) = \int_0^1 f + \int_0^1 g = 0 + 0 = 0$$

Therefore, $f + g \in W_0$.

3. **Closed under scalar multiplication:** Let $f \in W_0$ and $\lambda \in \mathbb{R}$. The scalar multiple λf is continuous, and by the linearity of the integral:

$$\int_0^1 (\lambda f) = \lambda \int_0^1 f = \lambda(0) = 0$$

Therefore, $\lambda f \in W_0$.

Thus, W_b is a subspace of $\mathbb{R}^{[0,1]}$ if and only if $b = 0$. □

Question 5. Is $\mathbb{R}^2$ a subspace of the complex vector space $\mathbb{C}^2$?

Proof. No, $\mathbb{R}^2$ is not a subspace of the complex vector space $\mathbb{C}^2$.

To be a subspace of the complex vector space $\mathbb{C}^2$, the subset $\mathbb{R}^2$ must be closed under scalar multiplication for all scalars in the field $\mathbb{C}$.

Consider the vector $v = (1, 0) \in \mathbb{R}^2$ and the complex scalar $\lambda = i \in \mathbb{C}$. Multiplying the vector by the scalar yields:

$$\lambda v = i(1, 0) = (i, 0)$$

Since $i \notin \mathbb{R}$, the resulting vector $(i, 0)$ is not in $\mathbb{R}^2$. □

Question 6. (a) Is $\{(a, b, c) \in \mathbb{R}^3 : a^3 = b^3\}$ a subspace of $\mathbb{R}^3$?
 (b) Is $\{(a, b, c) \in \mathbb{C}^3 : a^3 = b^3\}$ a subspace of $\mathbb{C}^3$?

Proof. **(a) Yes, it is a subspace.** Let $W = \{(a, b, c) \in \mathbb{R}^3 : a^3 = b^3\}$. For real numbers, the function $f(x) = x^3$ is injective, so $a^3 = b^3$ implies $a = b$. Thus, W can be rewritten as $W = \{(a, a, c) \mid a, c \in \mathbb{R}\}$. We verify the subspace conditions:

1. Additive identity: The zero vector $(0, 0, 0)$ is in W since $0 = 0$.
2. Closed under addition: Let $v_1 = (a_1, a_1, c_1)$ and $v_2 = (a_2, a_2, c_2)$ be in W . Their sum $v_1 + v_2 = (a_1 + a_2, a_1 + a_2, c_1 + c_2)$ is in W because its first two coordinates are equal.
3. Closed under scalar multiplication: Let $v = (a, a, c) \in W$ and $\lambda \in \mathbb{R}$. The scalar multiple $\lambda v = (\lambda a, \lambda a, \lambda c)$ is in W because its first two coordinates are equal.

Therefore, W is a subspace of $\mathbb{R}^3$.

(b) No, it is not a subspace. Consider the vectors $v_1 = (1, 1, 0)$ and $v_2 = (\omega, 1, 0)$. Both vectors are in U because $1^3 = 1^3$ and $\omega^3 = 1 = 1^3$. Their sum is:

$$v_1 + v_2 = (1 + \omega, 2, 0)$$

For this sum to be in U , its coordinates must satisfy $a^3 = b^3$, meaning we must have $(1 + \omega)^3 = 2^3 = 8$. Expanding the left side:

$$(1 + \omega)^3 = 1^3 + 3(1^2)\omega + 3(1)\omega^2 + \omega^3 = 1 + 3\omega + 3\omega^2 + 1 = 2 + 3(\omega + \omega^2)$$

Since $1 + \omega + \omega^2 = 0$ for cube roots of unity, we know $\omega + \omega^2 = -1$. Thus:

$$(1 + \omega)^3 = 2 + 3(-1) = -1$$

Since $-1 \neq 8$, the vector $v_1 + v_2 \notin U$. Because U is not closed under addition, it is not a subspace of $\mathbb{C}^3$. □

Question 7. Prove or give a counterexample: If U is a nonempty subset of $\mathbb{R}^2$ such that U is closed under addition and under taking additive inverses (meaning $-u \in U$ whenever $u \in U$), then U is a subspace of $\mathbb{R}^2$.

Proof. The statement is false. We provide a counterexample.

Let $U = \mathbb{Z}^2 = \{(m, n) \mid m, n \in \mathbb{Z}\}$, which is the set of all vectors in $\mathbb{R}^2$ with integer coordinates.

We verify that U satisfies the given conditions:

1. **U is nonempty:** The vector $(0, 0) \in U$ because 0 is an integer.
2. **Closed under addition:** Let $u = (m_1, n_1) \in U$ and $v = (m_2, n_2) \in U$. Their sum is $u + v = (m_1 + m_2, n_1 + n_2)$. Since the sum of two integers is an integer, $u + v \in U$.
3. **Closed under additive inverses:** Let $u = (m, n) \in U$. Its additive inverse is $-u = (-m, -n)$. Since the negative of an integer is an integer, $-u \in U$.

However, U is not a subspace of $\mathbb{R}^2$ because it is not closed under scalar multiplication over the field $\mathbb{R}$.

For a specific counterexample, let $u = (1, 1) \in U$ and let the scalar $\lambda = \frac{1}{2} \in \mathbb{R}$. Multiplying them yields:

$$\lambda u = \frac{1}{2}(1, 1) = \left(\frac{1}{2}, \frac{1}{2}\right)$$

Because $\frac{1}{2}$ is not an integer, the resulting vector $\left(\frac{1}{2}, \frac{1}{2}\right) \notin U$.

Since U is not closed under scalar multiplication, it fails the subspace criterion. Therefore, the statement is false. $\square$

Question 8. Give an example of a nonempty subset U of $\mathbb{R}^2$ such that U is closed under scalar multiplication, but U is not a subspace of $\mathbb{R}^2$.

Proof. $U = \{(x, y) \in \mathbb{R}^2 \mid x = 0 \text{ or } y = 0\}$ $\square$

Question 9. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called *periodic* if there exists a positive number p such that $f(x) = f(x + p)$ for all $x \in \mathbb{R}$. Is the set of periodic functions from $\mathbb{R}$ to $\mathbb{R}$ a subspace of $\mathbb{R}^{\mathbb{R}}$? Explain.

Proof. We provide a counterexample. Let $f(x) = \cos(x)$ and $g(x) = \cos(\pi x)$. Both f and g are periodic functions:

- f has a period of 2π , since $\cos(x + 2\pi) = \cos(x)$.
- g has a period of 2, since $\cos(\pi(x + 2)) = \cos(\pi x + 2\pi) = \cos(\pi x)$.

Let $h = f + g$, so $h(x) = \cos(x) + \cos(\pi x)$. We will show that h is not periodic. Suppose, for the sake of contradiction, that h is periodic with some positive period $p > 0$. Then $h(x) = h(x + p)$ for all $x \in \mathbb{R}$. Evaluating at $x = 0$, we get:

$$h(p) = h(0)$$

$$\cos(p) + \cos(\pi p) = \cos(0) + \cos(\pi \cdot 0) = 2$$

Since the maximum value of the real-valued cosine function is 1, the sum $\cos(p) + \cos(\pi p)$ can only equal 2 if both terms are exactly 1:

$$\cos(p) = 1 \quad \text{and} \quad \cos(\pi p) = 1$$

The condition $\cos(p) = 1$ implies that $p = 2\pi k$ for some positive integer k . The condition $\cos(\pi p) = 1$ implies that $\pi p = 2\pi m$, which simplifies to $p = 2m$ for some positive integer m .

Equating the two expressions for p , we have:

$$2\pi k = 2m \implies \pi = \frac{m}{k}$$

Because k and m are positive integers, this implies that π is a rational number. This is a contradiction, as π is known to be irrational.

Therefore, our assumption that h is periodic must be false. Because f and g are periodic but their sum $f + g$ is not, the set of periodic functions is not closed under addition. Consequently, it is not a subspace of $\mathbb{R}^{\mathbb{R}}$. $\square$

Question 10. Suppose $\mathcal{V}_1$ and $\mathcal{V}_2$ are subspaces of $\mathcal{V}$. Prove that the intersection $\mathcal{V}_1 \cap \mathcal{V}_2$ is a subspace of $\mathcal{V}$.

Proof. Let $\mathcal{V}_1$ and $\mathcal{V}_2$ be subspaces of a vector space $\mathcal{V}$ over a field $\mathbb{F}$.

1. **Additive identity:** Because $\mathcal{V}_1$ and $\mathcal{V}_2$ are subspaces of $\mathcal{V}$, both must contain the additive identity of $\mathcal{V}$. Thus, $0 \in \mathcal{V}_1$ and $0 \in \mathcal{V}_2$. Since 0 is in both sets, it is in their intersection, meaning $0 \in \mathcal{V}_1 \cap \mathcal{V}_2$.
2. **Closed under addition:** Let $u, w \in \mathcal{V}_1 \cap \mathcal{V}_2$. By definition of the intersection, $u, w \in \mathcal{V}_1$ and $u, w \in \mathcal{V}_2$. Because $\mathcal{V}_1$ is a subspace, it is closed under addition, so $u + w \in \mathcal{V}_1$. Similarly, because $\mathcal{V}_2$ is a subspace, $u + w \in \mathcal{V}_2$. Since $u + w$ is in both $\mathcal{V}_1$ and $\mathcal{V}_2$, it follows that $u + w \in \mathcal{V}_1 \cap \mathcal{V}_2$.
3. **Closed under scalar multiplication:** Let $u \in \mathcal{V}_1 \cap \mathcal{V}_2$ and let $\lambda \in \mathbb{F}$. By definition of the intersection, $u \in \mathcal{V}_1$ and $u \in \mathcal{V}_2$. Because $\mathcal{V}_1$ and $\mathcal{V}_2$ are subspaces, both are closed under scalar multiplication. Therefore, $\lambda u \in \mathcal{V}_1$ and $\lambda u \in \mathcal{V}_2$. Since λu is in both sets, $\lambda u \in \mathcal{V}_1 \cap \mathcal{V}_2$.

$\square$

Question 11. Prove that the intersection of every collection of subspaces of V is a subspace of V .

Proof. Let $\{V_i\}_{i \in I}$ be a collection of subspaces of a vector space V over a field $\mathbb{F}$, where I is an index set. Let $W = \bigcap_{i \in I} V_i$ be the intersection of this collection. We verify that W satisfies the three conditions of a subspace:

1. **Additive identity:** Since each V_i is a subspace of $\mathcal{V}$, the additive identity 0 must be in every V_i for all $i \in I$. Because 0 is in every subspace in the collection, it is in their intersection. Thus, $0 \in W$.
2. **Closed under addition:** Let $u, w \in W$. By the definition of the intersection, $u \in V_i$ and $w \in V_i$ for every $i \in I$. Because every V_i is a subspace, they are all individually closed under addition, meaning $u + w \in V_i$ for all $i \in I$. Since $u + w$ is in every subspace in the collection, $u + w \in \bigcap_{i \in I} V_i = W$.
3. **Closed under scalar multiplication:** Let $u \in W$ and $\lambda \in \mathbb{F}$. By definition, $u \in V_i$ for all $i \in I$. Because each V_i is a subspace, it is closed under scalar multiplication, so $\lambda u \in V_i$ for all $i \in I$. Since λu is in every subspace in the collection, $\lambda u \in \bigcap_{i \in I} V_i = W$.

Since the intersection W contains the additive identity and is closed under both addition and scalar multiplication, it is a subspace of $\mathcal{V}$. $\square$

Question 12. Prove that the union of two subspaces of $\mathcal{V}$ is a subspace of $\mathcal{V}$ if and only if one of the subspaces is contained in the other.

Proof. Let V_1 and V_2 be subspaces of a vector space V . We must prove that $V_1 \cup V_2$ is a subspace if and only if $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$.

($\Leftarrow$) Assume one of the subspaces is contained in the other. Without loss of generality, let $V_1 \subseteq V_2$. Then their union is $V_1 \cup V_2 = V_2$. Since V_2 is given to be a subspace of V , it follows immediately that $V_1 \cup V_2$ is a subspace of V .

($\Rightarrow$) Assume that the union $V_1 \cup V_2$ is a subspace of V . We want to show that either $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$.

Suppose, for the sake of contradiction, that neither subspace is contained in the other. This means there exists some vector $u \in V_1$ such that $u \notin V_2$, and there exists some vector $w \in V_2$ such that $w \notin V_1$.

Since $u \in V_1$ and $w \in V_2$, both vectors are in $V_1 \cup V_2$. Because we assumed $V_1 \cup V_2$ is a subspace, it must be closed under vector addition. Therefore, the sum $u + w$ must be in $V_1 \cup V_2$. By the definition of the union, this implies that either $u + w \in V_1$ or $u + w \in V_2$.

Case 1: Suppose $u + w \in V_1$. Since V_1 is a subspace, it is closed under additive inverses and addition, so it must contain $-u$. Therefore, $(u + w) + (-u) \in V_1$, which simplifies to $w \in V_1$. This directly contradicts our initial assumption that $w \notin V_1$.

Case 2: Suppose $u + w \in V_2$. Similarly, since V_2 is a subspace, it contains $-w$. Therefore, $(u + w) + (-w) \in V_2$, which simplifies to $u \in V_2$. This directly contradicts our initial assumption that $u \notin V_2$.

In both cases, we arrive at a contradiction. Therefore, our assumption that neither subspace is contained in the other must be false. We conclude that $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$. $\square$

Question 13. Prove that the union of three subspaces of $\mathcal{V}$ is a subspace of $\mathcal{V}$ if and only if one of the subspaces contains the other two.

Proof. Let $W = \mathcal{V}_1 \cup \mathcal{V}_2 \cup \mathcal{V}_3$. We want to prove that W is a subspace of $\mathcal{V}$ if and only if one of the subspaces contains the other two. (Note: This proof assumes the underlying field $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$, which means it has more than 2 elements).

($\Leftarrow$) Assume one subspace contains the other two. Without loss of generality, let $\mathcal{V}_1$ contain $\mathcal{V}_2$ and $\mathcal{V}_3$. Then the union $\mathcal{V}_1 \cup \mathcal{V}_2 \cup \mathcal{V}_3 = \mathcal{V}_1$. Because $\mathcal{V}_1$ is given to be a subspace of $\mathcal{V}$, their union is trivially a subspace of $\mathcal{V}$.

($\Rightarrow$) Assume $W = \mathcal{V}_1 \cup \mathcal{V}_2 \cup \mathcal{V}_3$ is a subspace of $\mathcal{V}$. We want to show that one of the subspaces contains the other two.

Suppose for the sake of contradiction that no single subspace contains the other two. If the union of any two of these subspaces completely covered W (for example, if $W = \mathcal{V}_1 \cup \mathcal{V}_2$), then by the known theorem for the union of two subspaces, we would have $\mathcal{V}_1 \subseteq \mathcal{V}_2$ or $\mathcal{V}_2 \subseteq \mathcal{V}_1$. This would mean $W = \mathcal{V}_2$ or $W = \mathcal{V}_1$, implying that one subspace contains all three. Since we assumed no single subspace contains the other two, no two subspaces can cover W .

Therefore, there must exist vectors that strictly belong to only one of the subspaces:

- There is some $v_1 \in \mathcal{V}_1$ such that $v_1 \notin \mathcal{V}_2 \cup \mathcal{V}_3$.
- There is some $v_2 \in \mathcal{V}_2$ such that $v_2 \notin \mathcal{V}_1 \cup \mathcal{V}_3$.

Because W is a subspace, it is closed under vector addition and scalar multiplication. Let c be any non-zero scalar. The vector $v_1 + cv_2$ must be in W , meaning it is in $\mathcal{V}_1$, $\mathcal{V}_2$, or $\mathcal{V}_3$.

- If $v_1 + cv_2 \in \mathcal{V}_1$: Since $\mathcal{V}_1$ is a subspace, it contains $-v_1$. Thus, $(v_1 + cv_2) - v_1 = cv_2 \in \mathcal{V}_1$. Because $c \neq 0$, this implies $v_2 \in \mathcal{V}_1$, which contradicts our choice of v_2 .
- If $v_1 + cv_2 \in \mathcal{V}_2$: Since $\mathcal{V}_2$ is a subspace, it contains $-cv_2$. Thus, $(v_1 + cv_2) - cv_2 = v_1 \in \mathcal{V}_2$, which contradicts our choice of v_1 .

Thus, for every non-zero scalar c , the vector $v_1 + cv_2$ must belong to $\mathcal{V}_3$.

Because our field $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$, there are infinitely many scalars. We can pick two distinct non-zero scalars, say $c = 1$ and $c = 2$. Based on the logic above, both $v_1 + v_2 \in \mathcal{V}_3$ and $v_1 + 2v_2 \in \mathcal{V}_3$. Since $\mathcal{V}_3$ is a subspace, it is closed under vector subtraction, meaning:

$$(v_1 + 2v_2) - (v_1 + v_2) = v_2 \in \mathcal{V}_3$$

But this directly contradicts our initial choice that $v_2 \notin \mathcal{V}_1 \cup \mathcal{V}_3$.

This contradiction means our assumption that no subspace contains the other two must be false. Therefore, one of the subspaces must contain the other two. $\square$

Question 14. Suppose

$$U = \{(x, -x, 2x) \in \mathbb{F}^3 : x \in \mathbb{F}\} \quad \text{and} \quad W = \{(x, x, 2x) \in \mathbb{F}^3 : x \in \mathbb{F}\}.$$

Describe $U + W$ using symbols, and also give a description of $U + W$ that uses no symbols.

Proof. Let $u = (x, -x, 2x) \in U$ and $w = (y, y, 2y) \in W$ for some scalars $x, y \in \mathbb{F}$. Their sum is:

$$u + w = (x, -x, 2x) + (y, y, 2y) = (x + y, -x + y, 2x + 2y)$$

Let this resulting vector be (a, b, c) . We can relate its coordinates to the scalars x and y :

$$\begin{aligned} a &= x + y \\ b &= -x + y \\ c &= 2x + 2y = 2(x + y) \end{aligned}$$

Notice that the third coordinate c is always exactly twice the first coordinate a (since $c = 2a$).

To ensure that every vector of the form $(a, b, 2a)$ is indeed in $U + W$, we must show that for any given $a, b \in \mathbb{F}$, there exist $x, y \in \mathbb{F}$ that satisfy the system:

$$\begin{aligned} x + y &= a \\ -x + y &= b \end{aligned}$$

So $y = \frac{a+b}{2}$, $x = \frac{a-b}{2}$. Because the underlying field $\mathbb{F}$ is either $\mathbb{R}$ or $\mathbb{C}$, division by 2 is valid, meaning such scalars x and y will always exist for any choice of a and b .

Therefore, the sum $U + W$ can be described using symbols as:

$$U + W = \{(a, b, 2a) \in \mathbb{F}^3 : a, b \in \mathbb{F}\}$$

Alternatively, writing the condition as a relationship between coordinates (x, y, z) :

$$U + W = \{(x, y, z) \in \mathbb{F}^3 : z = 2x\}$$

Description using no symbols:

The sum is the set of all vectors in the three-dimensional vector space where the third coordinate is exactly twice the first coordinate. $\square$

Question 15. Suppose U is a subspace of V . What is $U + U$?

Proof. We claim that $U + U = U$.

To prove this, we show subset inclusion in both directions.

First, we show that $U + U \subseteq U$. Let $x \in U + U$. By definition of the sum of subsets, $x = u_1 + u_2$ for some vectors $u_1, u_2 \in U$. Since U is a subspace, it is closed under vector addition, which implies $u_1 + u_2 \in U$. Therefore, $x \in U$, establishing that $U + U \subseteq U$.

Next, we show that $U \subseteq U + U$. Let $u \in U$. Since U is a subspace, it contains the additive identity 0. We can express u as the sum $u = u + 0$. Because $u \in U$ and $0 \in U$, the sum $u + 0$ is an element of $U + U$. Therefore, $u \in U + U$, establishing that $U \subseteq U + U$.

Since $U + U \subseteq U$ and $U \subseteq U + U$, we conclude that $U + U = U$. $\square$

Question 16. Is the operation of addition on the subspaces of $\mathcal{V}$ commutative? In other words, if U and W are subspaces of $\mathcal{V}$, is $U + W = W + U$?

Proof. Yes, the operation of addition on the subspaces of $\mathcal{V}$ is commutative.

To prove that $U + W = W + U$, we must show that each set is a subset of the other.

First, let $x \in U + W$. By the definition of the sum of subspaces, $x = u + w$ for some vectors $u \in U$ and $w \in W$. Because vector addition in V is commutative, we have $u + w = w + u$. Since $w \in W$ and $u \in U$, the vector $w + u$ is in $W + U$. Therefore, $x \in W + U$, establishing that $U + W \subseteq W + U$.

Next, let $y \in W + U$. By definition, $y = w + u$ for some vectors $w \in W$ and $u \in U$. Again, by the commutativity of vector addition in V , $w + u = u + w$. Since $u \in U$ and $w \in W$, the vector $u + w$ is in $U + W$. Therefore, $y \in U + W$, establishing that $W + U \subseteq U + W$.

Because $U + W \subseteq W + U$ and $W + U \subseteq U + W$, we conclude that $U + W = W + U$. $\square$

Question 17. Is the operation of addition on the subspaces of $\mathcal{V}$ associative? In other words, if $\mathcal{V}_1, \mathcal{V}_2, \mathcal{V}_3$ are subspaces of $\mathcal{V}$, is

$$(\mathcal{V}_1 + \mathcal{V}_2) + \mathcal{V}_3 = \mathcal{V}_1 + (\mathcal{V}_2 + \mathcal{V}_3)?$$

Proof. Yes, the operation of addition on the subspaces of $\mathcal{V}$ is associative.

To prove that $(\mathcal{V}_1 + \mathcal{V}_2) + \mathcal{V}_3 = \mathcal{V}_1 + (\mathcal{V}_2 + \mathcal{V}_3)$, we show subset inclusion in both directions.

First, let $x \in (\mathcal{V}_1 + \mathcal{V}_2) + \mathcal{V}_3$. By definition, $x = u + v_3$ for some $u \in \mathcal{V}_1 + \mathcal{V}_2$ and $v_3 \in \mathcal{V}_3$. Since $u \in \mathcal{V}_1 + \mathcal{V}_2$, we can write $u = v_1 + v_2$ for some $v_1 \in \mathcal{V}_1$ and $v_2 \in \mathcal{V}_2$. Substituting for u yields $x = (v_1 + v_2) + v_3$. Because vector addition in V is associative, we can rewrite this as $x = v_1 + (v_2 + v_3)$. Let $w = v_2 + v_3$. Since $v_2 \in \mathcal{V}_2$ and $v_3 \in \mathcal{V}_3$, by definition $w \in \mathcal{V}_2 + \mathcal{V}_3$. Therefore, $x = v_1 + w$, which means $x \in \mathcal{V}_1 + (\mathcal{V}_2 + \mathcal{V}_3)$. This establishes that $(\mathcal{V}_1 + \mathcal{V}_2) + \mathcal{V}_3 \subseteq \mathcal{V}_1 + (\mathcal{V}_2 + \mathcal{V}_3)$.

Next, let $y \in \mathcal{V}_1 + (\mathcal{V}_2 + \mathcal{V}_3)$. By definition, $y = v_1 + w$ for some $v_1 \in \mathcal{V}_1$ and $w \in \mathcal{V}_2 + \mathcal{V}_3$. Since $w \in \mathcal{V}_2 + \mathcal{V}_3$, we can write $w = v_2 + v_3$ for some $v_2 \in \mathcal{V}_2$ and $v_3 \in \mathcal{V}_3$. Substituting for w yields $y = v_1 + (v_2 + v_3)$. By the associativity of vector addition in $\mathcal{V}$, we can rewrite this as $y = (v_1 + v_2) + v_3$. Let $u = v_1 + v_2$. Since $v_1 \in \mathcal{V}_1$ and $v_2 \in \mathcal{V}_2$, by definition $u \in \mathcal{V}_1 + \mathcal{V}_2$. Therefore, $y = u + v_3$, which means $y \in (\mathcal{V}_1 + \mathcal{V}_2) + \mathcal{V}_3$. This establishes that $\mathcal{V}_1 + (\mathcal{V}_2 + \mathcal{V}_3) \subseteq (\mathcal{V}_1 + \mathcal{V}_2) + \mathcal{V}_3$.

Since both sets are subsets of each other, $(\mathcal{V}_1 + \mathcal{V}_2) + \mathcal{V}_3 = \mathcal{V}_1 + (\mathcal{V}_2 + \mathcal{V}_3)$. $\square$

Question 18. Does the operation of addition on the subspaces of $\mathcal{V}$ have an additive identity? Which subspaces have additive inverses?

Proof. Yes, the operation of addition on the subspaces of $\mathcal{V}$ has an additive identity, which is the trivial subspace $\{0\}$.

To verify this, let U be any subspace of V . By the definition of subspace addition,

$$U + \{0\} = \{u + 0 \mid u \in U\} = \{u \mid u \in U\} = U$$

Therefore, $\{0\}$ is the additive identity.

Regarding additive inverses, only the trivial subspace $\{0\}$ has an additive inverse.

To prove this, suppose a subspace U has an additive inverse W . By definition, this means $U + W = \{0\}$. Because W is a subspace, it must contain the zero vector 0 . Let u be any vector in U . The sum $u + 0$ must be in $U + W$. Since $U + W = \{0\}$, it follows that $u + 0 = 0$, which means $u = 0$. Because this must be true for every vector $u \in U$, the only element in U is the zero vector, so $U = \{0\}$. Thus, no non-trivial subspace can have an additive inverse. $\square$

Question 19. Prove or give a counterexample: If $\mathcal{V}_1, \mathcal{V}_2, U$ are subspaces of $\mathcal{V}$ such that

$$\mathcal{V}_1 + U = \mathcal{V}_2 + U,$$

then $\mathcal{V}_1 = \mathcal{V}_2$.

Proof. The statement is false. We provide a counterexample.

Let $V = \mathbb{R}^2$. Consider the following subspaces:

- $U = \{(x, 0) \mid x \in \mathbb{R}\}$ (the x -axis)
- $V_1 = \{(0, y) \mid y \in \mathbb{R}\}$ (the y -axis)
- $V_2 = \{(x, x) \mid x \in \mathbb{R}\}$ (the line $y = x$)

First, we evaluate $\mathcal{V}_1 + U$. Any vector $(a, b) \in \mathbb{R}^2$ can be written as $(0, b) + (a, 0)$. Since $(0, b) \in \mathcal{V}_1$ and $(a, 0) \in U$, their sum spans the entire space. Thus, $\mathcal{V}_1 + U = \mathbb{R}^2$.

Next, we evaluate $\mathcal{V}_2 + U$. Any vector $(a, b) \in \mathbb{R}^2$ can be written as $(b, b) + (a - b, 0)$. Since $(b, b) \in \mathcal{V}_2$ and $(a - b, 0) \in U$, this sum also spans the entire space. Thus, $\mathcal{V}_2 + U = \mathbb{R}^2$.

We have established that $\mathcal{V}_1 + U = \mathcal{V}_2 + U = \mathbb{R}^2$. However, $\mathcal{V}_1 \neq \mathcal{V}_2$ (for example, the vector $(0, 1)$ is in $\mathcal{V}_1$ but not in $\mathcal{V}_2$). Because $\mathcal{V}_1 + U = \mathcal{V}_2 + U$ does not imply $\mathcal{V}_1 = \mathcal{V}_2$, the statement is false. $\square$

Question 20. Suppose

$$U = \{(x, x, y, y) \in \mathbb{F}^4 : x, y \in \mathbb{F}\}.$$

Find a subspace W of $\mathbb{F}^4$ such that $\mathbb{F}^4 = U \oplus W$.

Proof. $W = \{(0, a, 0, b) \in \mathbb{F}^4 : a, b \in \mathbb{F}\}$. $\square$

Question 21. Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbb{F}^5 : x, y \in \mathbb{F}\}.$$

Find a subspace W of $\mathbb{F}^5$ such that $\mathbb{F}^5 = U \oplus W$.

Proof. $W = \{(0, 0, a, b, c) \in \mathbb{F}^5 : a, b, c \in \mathbb{F}\}$. $\square$

Question 22. Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbb{F}^5 : x, y \in \mathbb{F}\}.$$

Find three subspaces W_1, W_2, W_3 of $\mathbb{F}^5$, none of which equals $\{0\}$, such that $\mathbb{F}^5 = U \oplus W_1 \oplus W_2 \oplus W_3$.

Proof.

$$W_1 = \{(0, 0, a, 0, 0) \in \mathbb{F}^5 : a \in \mathbb{F}\}$$

$$W_2 = \{(0, 0, 0, b, 0) \in \mathbb{F}^5 : b \in \mathbb{F}\}$$

$$W_3 = \{(0, 0, 0, 0, c) \in \mathbb{F}^5 : c \in \mathbb{F}\}$$

$\square$

Question 23. Prove or give a counterexample: If $\mathcal{V}_1, \mathcal{V}_2, U$ are subspaces of $\mathcal{V}$ such that

$$\mathcal{V} = \mathcal{V}_1 \oplus U \quad \text{and} \quad \mathcal{V} = \mathcal{V}_2 \oplus U,$$

then $\mathcal{V}_1 = \mathcal{V}_2$.

Proof. The statement is false. We provide a counterexample.

- $U = \{(x, 0) \mid x \in \mathbb{R}\}$ (the x -axis)
- $\mathcal{V}_1 = \{(0, y) \mid y \in \mathbb{R}\}$ (the y -axis)

- $\mathcal{V}_2 = \{(y, y) \mid y \in \mathbb{R}\}$ (the line $y = x$)

□

Question 24. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called *even* if

$$f(-x) = f(x)$$

for all $x \in \mathbb{R}$. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called *odd* if

$$f(-x) = -f(x)$$

for all $x \in \mathbb{R}$. Let $\mathcal{V}_e$ denote the set of real-valued even functions on $\mathbb{R}$ and let $\mathcal{V}_o$ denote the set of real-valued odd functions on $\mathbb{R}$. Show that $\mathbb{R}^{\mathbb{R}} = \mathcal{V}_e \oplus \mathcal{V}_o$

Proof. To prove that $\mathbb{R}^{\mathbb{R}} = \mathcal{V}_e \oplus \mathcal{V}_o$, we must show that $\mathcal{V}_e \cap \mathcal{V}_o = \{0\}$ and $\mathcal{V}_e + \mathcal{V}_o = \mathbb{R}^{\mathbb{R}}$.

First, we show that $\mathcal{V}_e \cap \mathcal{V}_o = \{0\}$. Let $f \in \mathcal{V}_e \cap \mathcal{V}_o$. Because $f \in \mathcal{V}_e$, it must be an even function, so $f(-x) = f(x)$ for all $x \in \mathbb{R}$. Because $f \in \mathcal{V}_o$, it must be an odd function, so $f(-x) = -f(x)$ for all $x \in \mathbb{R}$. Equating the two expressions for $f(-x)$, we get:

$$f(x) = -f(x)$$

$$2f(x) = 0$$

$$f(x) = 0$$

Since $f(x) = 0$ for all $x \in \mathbb{R}$, f is the zero function. Thus, $\mathcal{V}_e \cap \mathcal{V}_o = \{0\}$.

Next, we show that $\mathcal{V}_e + \mathcal{V}_o = \mathbb{R}^{\mathbb{R}}$. Let $f \in \mathbb{R}^{\mathbb{R}}$ be an arbitrary real-valued function. We can rewrite $f(x)$ identically as:

$$f(x) = \frac{f(x) + f(-x)}{2} + \frac{f(x) - f(-x)}{2}$$

Let $f_e(x) = \frac{f(x) + f(-x)}{2}$ and $f_o(x) = \frac{f(x) - f(-x)}{2}$. We now check if $f_e \in \mathcal{V}_e$ and $f_o \in \mathcal{V}_o$. For f_e :

$$f_e(-x) = \frac{f(-x) + f(-(-x))}{2} = \frac{f(-x) + f(x)}{2} = f_e(x)$$

Since $f_e(-x) = f_e(x)$, f_e is an even function, so $f_e \in \mathcal{V}_e$. For f_o :

$$f_o(-x) = \frac{f(-x) - f(-(-x))}{2} = \frac{f(-x) - f(x)}{2} = -\left(\frac{f(x) - f(-x)}{2}\right) = -f_o(x)$$

Since $f_o(-x) = -f_o(x)$, f_o is an odd function, so $f_o \in \mathcal{V}_o$.

Since any arbitrary function $f \in \mathbb{R}^{\mathbb{R}}$ can be expressed as the sum of an even function $f_e \in \mathcal{V}_e$ and an odd function $f_o \in \mathcal{V}_o$, we have $\mathcal{V}_e + \mathcal{V}_o = \mathbb{R}^{\mathbb{R}}$.

Since $\mathcal{V}_e \cap \mathcal{V}_o = \{0\}$ and $\mathcal{V}_e + \mathcal{V}_o = \mathbb{R}^{\mathbb{R}}$, we conclude that $\mathbb{R}^{\mathbb{R}} = \mathcal{V}_e \oplus \mathcal{V}_o$. □